Propose, Do Not Certify: Representation Quality and the Judge/Proposer Gap

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Abstract

A representation that is an unreliable judge can still be a useful proposer — but less decisively than a random-split evaluation suggests. On a 582-entry, 57-cell ontology filing task, embedding geometry achieves top-3 containment of the rater's own cell at 0.668 under a random train/test split against a matched 1,000-shuffle permuted-label null of 0.159. Under a *sequential* split — first N decisions train, later decisions test, which is what a real user experiences — top-3 falls to 0.541, the learning curve is no longer monotone, and the advantage over the random split is about thirteen points.

We deliberately do *not* subtract the proposer score from the classifier score. Containment in $\{0, 1\}$ and ROC AUC are different metrics on different scales, and their difference has no interpretation; an earlier draft of this paper reported such a "gap" ($0.668 - 0.519$) and that arithmetic is withdrawn here. Each task is instead reported against its own matched null. Judged that way: as a proposer the representation clears its random-split permuted null by $+0.509$ top-3, and as a certifier it sits at chance (0.519 against a 0.5 null, $n = 574$). The surviving claim is therefore qualitative and about decision leverage — the representation narrows 57 candidate cells to 3 and places the right one in that set 54–67% of the time, whereas the same representation asked to certify is uninformative — not a numerical gap between two incommensurable scales.

1 Introduction

Every refutation in the Physis research track refuted unsupervised, zero-shot judgement. COSINE, NAME-ONLY, CONSTRAINT, ANCESTRY and ANCHORVOC are fixed formulas with nothing fitted to a human decision, and on real misfilings they sit at chance (0.508–0.536). The supervised, human-in-the-loop version was never measured — and the question a person filing 582 entries has is not "is the machine right?" but "can it narrow 57 cells to 3 so I choose instead of search?"

The proposition we test is not that geometry can judge — Section 5 shows it cannot — but that it can propose: the representation that cannot judge may still usefully narrow a search. We measure that separately, in each case against the null appropriate to the task, and find the proposer's advantage real under both a random and a sequential split while the certifier's is absent under both. We do not report a distance between the two tasks, for the reasons given in Section 7.

2 Method

Ground truth is the rater's *revised* cell in `refiling_gate0.tsv`: 582 blind re-filings, 57 distinct target cells. The 76 non-blind entries are excluded. Vectors are the shipped bge-base-en-v1.5 768-d from `corpus.json`; no inference is run.

Nearest-centroid over cells built from the decisions already made, scored as top-k containment of the rater's own cell, on a learning curve over how many decisions the rater has made. Five shuffles per point.

3 Results

Decisions seen	Top-1	Top-3	Top-5
58	0.211	0.385	0.465
145	0.307	0.508	0.591
291	0.368	0.595	0.689
436	**0.438**	**0.667**	**0.755**

Table 1: Top-k containment of the rater's own cell, by number of decisions seen. The learning curve is still climbing at 436.

Controls at 436 decisions seen:

Arm	Top-1	Top-3	Top-5
GEOMETRY	**0.438**	**0.667**	**0.755**
PRIOR old-¿new lookup	0.184	0.368	0.474
MAJORITY	0.089	0.255	0.359
PERMUTED (null)	0.036	0.136	0.233
LEN-ONLY	0.010	0.036	0.071

Table 2: Controls at 436 decisions, all arms at five random splits so the comparison is internally consistent. Geometry beats its permuted null by +0.531 top-3, and beats the lookup table by +0.299. The null row is the weakest number here — see the text for the 1,000-shuffle re-estimate.

Geometry beats its permuted null by +0.531 top-3 at this seed count, and beats the lookup table by +0.299. The learning curve appears to be still climbing at 436 under this split — but that claim does not survive the sequential split reported below, where the curve falls at the last step, so we withdraw it.

One row of that table has since been re-estimated. The permuted-null arm averaged only five shuffles, which is too few for a permutation baseline, so it was re-run with 1,000 shuffles at the same split. That raises the null from 0.136 to 0.159 top-3, which narrows the geometry-minus-null gap from +0.531 to +0.509. Geometry under 1,000 splits reads 0.419 / 0.668 / 0.769, against the 0.438 / 0.667 / 0.755 shown above, so the two estimates agree to within about 0.02 at every k. We report both rather than silently mixing seed counts within one table, and we take +0.509 as the defensible gap.

4 Discussion

Each task against its own null — not against each other. The tempting summary of this result is a single number: proposer top-3 minus certifier AUC. We do not report one, because the two quantities are not commensurable. Top-k containment is a $\{0, 1\}$ hit rate with a finite number of candidate cells; ROC AUC is a rank statistic over a threshold-free continuum. Neither the

difference $0.668 - 0.519$ nor the ratio $0.668/0.519$ has an interpretation — changing the candidate-set size from 3 to 10 would move the first term while leaving the second untouched. An earlier draft of this paper reported such a "gap" as the headline finding and computed its shrinkage across splits ($+0.149 \rightarrow +0.022$). That arithmetic is withdrawn.

The comparison is instead made within each task. As a proposer, geometry clears its matched permuted-label null by +0.509 top-3 under the random split (0.668 vs. 0.159). As a certifier, it clears its own null — chance, AUC 0.5 — by 0.019, which is not distinguishable from no improvement ($n = 574$). So the asymmetry is real but it is an asymmetry of *which null the task clears*, not a distance between two scores on different rulers.

The sequential protocol does not yet have its own permuted null: the sequential run measured the geometry arm and the in-order baselines but not a shuffled-label arm, so we report the sequential proposer score against the random-split null only as a directional observation and do not quote a sequential margin. Adding that arm is the first item in the experiment plan.

Put in a common, decision-relevant currency: the proposer takes a 57-cell search and hands a human 3 candidates containing the correct one 54–67% of the time (sequential to random), while the certifier, asked to answer the same question without a human in the loop, provides no usable signal at all. That is a statement about human decision leverage, and it is the claim we are prepared to defend.

The magnitude is split-dependent. The proposer's top-3 falls from 0.668 to 0.541 when evaluation becomes sequential — about thirteen points — which is the more realistic protocol, so the sequential figure is the conservative one and the random split is a labelled upper bound. What survives both splits is the ordering, not the size: geometry beats the lookup prior, the majority baseline and the length-only baseline at every fraction under both protocols, and the certifier stays at chance under both.

The earlier framing, that a number previously filed as a failure (top-1 0.447) becomes useful once the output is three candidates, still holds under both splits: top-3 exceeds top-1 by a factor of about 1.8 even sequentially, and a person choosing among three candidates has less work than a person searching 57 cells.

Supervision is doing the work. 436 decisions is most of a corpus. The honest claim is "geometry converts human decisions into leverage over the remaining ones," not "geometry replaces them." The learning curve is the thing to read: at 58 decisions top-3 is 0.385, already $1.5\times the majority floor$.

The mode axis is the constraint. Scoring per axis at 436 decisions seen:

Axis	Classes	Top-1	Top-2	Lift over majority
Domain	5	0.626	0.837	3.05×
Mode	14	0.460	0.678	1.97×

Table 3: Per-axis proposer performance. The machine can carry the domain axis and cannot carry the mode axis.

The machine can carry the domain axis and cannot carry the mode axis — 0.837 top-2 over five domains against 0.678 over fourteen modes, and the gap survives normalising for class count. This is the same asymmetry E6/E7 found and explained by population rather than by missing definition, now stated as a division of labour: **the machine proposes the domain, the person picks the mode.**

Limits. One rater (Claude), and the same rater produced the replacement filing. Under "is the machine right" that is a confound. Under this framing it is the target — the artifact exists

to pre-sort for an arbiter — but *whose* policy it learned is untested, and a second rater is the experiment that would settle it.

What this does not show. (1) Top-1 is not new — 0.438 essentially reproduces E5's 0.447 accuracy on the old labels. (2) It is not certification — nothing here says an entry belongs in the proposed cell. E14–E18 stand untouched. (3) It predicts a policy, not a truth. (4) The rater was a language model, so this measures whether geometry can predict one model's filing policy, not ontology truth. A second, independent human rater is the experiment that would settle it.

The split protocol, stated precisely

A learning-curve claim is only as good as its split, so we state ours exactly. At each fraction $f \in \{0.10, 0.25, 0.50, 0.75\}$ the 582 entries are shuffled and cut into a training prefix of $\max(2, \lfloor f \cdot 582 \rfloor)$ decisions and the remaining untouched decisions as test. Centroids are built from **train only**; test entries never contribute to a centroid. Each point is the mean over **five** random splits, and the reported k-seen values are those fractions (58, 145, 291, 436).

Two properties of this protocol are weaknesses, and we name them rather than leave them to be found:

- **The split is random in the main result; a sequential split was run in this revision.** A random split lets a test entry sit between training entries rather than after them, which is not what a real user experiences. We therefore ran the same protocol in file order — first N decisions as training, every later decision as test — and the difference is large. Reporting the sequential results as the more realistic estimate:

frac	train	test	top-1	top-3	top-5	prior-3	maj-3
0.10	58	524	0.195	0.345	0.422	0.250	0.218
0.25	145	437	0.195	0.348	0.430	0.272	0.224
0.50	291	291	0.323	0.577	0.677	0.309	0.230
0.75	436	146	0.308	0.541	0.644	0.267	0.123

  Against the random split at frac 0.75 (top-3 0.667–0.668), the sequential split gives 0.541 — the random split inflates top-3 by roughly +0.13. Two further consequences follow. First, the learning curve is *not* monotone in sequence: it rises from 0.345 to 0.577 and then *falls* to 0.541, so the earlier claim that "the curve is still climbing at 436" does not hold under this split. Second, at small N the geometry arm (0.345, 0.348) is barely distinguishable from the old→new lookup prior (0.250, 0.272), though it still comfortably beats majority (0.218, 0.224).

  Caveat on what this split is: `refiling_gate0.tsv` carries a sequence column but no timestamps, so we cannot confirm that the file order is the rater's adjudication order rather than the corpus order. This is a *sequential* split and we do not claim it is chronological. It bounds the random split's optimism; it does not establish the true temporal curve.

- **The permutation null was re-run with 1,000 shuffles.** The original arm averaged five shuffles, which our own project notes elsewhere call too weak for a permutation baseline. Re-running the null with 1,000 shuffles at the same split (0.75, 436 decisions seen) gives top-3 null 0.159 against the published 0.136. The five-shuffle figure was low by 0.023; we have no interval for it, since the original script reported only the mean over five, so we cannot say whether the

two differ significantly — only that the better-estimated null is higher. Corrected numbers at 436 decisions: null top-1 0.048, top-3 0.159, top-5 0.265; geometry top-1 0.419, top-3 0.668, top-5 0.769. The geometry-minus-null gap is therefore +0.371, +0.509 and +0.505 rather than the +0.576 implied by the original five-shuffle null. We report the revised figures and the reduced gap. The conclusion is unchanged in direction — geometry clears the permuted null by +0.509 top-3, while the strongest baseline over that same null, the old→new lookup prior, reaches only +0.209 (41% as much) and the length-only baseline does not clear the null at all (−0.123) — but the published null understated the noise, and we prefer to correct it ourselves than have a reviewer find it.

The revised null is stored in `research/e21_permutation_null_1000.json`; the re-run script is `research/papers/e21_perm_null_1000.py`, which imports the original `run_e21.py` so both arms come from the same code path.

Neither the remaining temporal-split gap nor any other weakness invalidates the direction of the result — the geometry arm beats every baseline by a wide margin — but they bound how much precision we claim.

Reproducibility and provenance

The E21 numbers above (top-k containment, the four baselines, the per-axis split) were regenerated from the surviving script and ground truth and are stored in `research/e21_proposer_results.json`; `research/papers/parse_e21.py` rebuilds that file and asserts the two headline figures.

The certification figures quoted for context — the 0.519 (COSINE) and 0.536 (CONSTRAINT) AUCs on real misfilings — come from the E15–E18 re-run stored in `research/e15_e18_rerun.json`. Those four experiments had shipped only as printing scripts with no saved output; they have since been re-run from the surviving corpus and ground truth, and reproduced their originally reported values (COSINE 0.882 under random injection and 0.519 on real misfilings; CONSTRAINT 0.612 and 0.536). The E5 figure 0.447 and the E24 figure 0.712 remain *not* independently reproducible: neither experiment has a saved result file in the working repository or the retired archive. They are quoted from the original reports and should be re-run before being relied on.